



ROOT CAUSE ANALYSIS

VIRTUAL CONTROL

VMWARE CLOUD FOUNDATION SOLUTIONS

RCA — VCF Appliance Root-Password Expiry & SSH Lockout

An expired appliance root password locked out SSH on VCF Operations (.77) — presenting as a wrong password. Root-caused, fixed, and scoped to the fleet: Fleet/LCM root expires ~Aug 1 (proactive window), vCenter root Aug 10. Management access only; no service impact.

v1.1

VMware Cloud Foundation 9.0.1

RCA — VCF Appliance Root-Password Expiry & SSH Lockout

Prepared by: Virtual Control LLC **Date:** July 27, 2026 **Environment:** VMware Cloud Foundation 9.0.1 — management appliances (VCF Operations, LCM/Fleet, vCenter, SDDC Manager)

Current status — RECOVERED (one appliance) / EXPOSURE OPEN (fleet). The VCF Operations appliance (.77) lost SSH access on 2026-07-26/27 due to an **expired root password** (not a wrong one) and was **restored**. A VCF Operations alert indicates **at least one other appliance root is on the same expiry path** (see §8). No workload or service outage occurred — this was a **management-access** incident only.

1. Incident Summary

- **What happened:** SSH to `root@192.168.1.77 (vcf-ops.lab.local)` failed with `Configured password was not accepted` — which reads like a *wrong* password, but the password value was correct. The account's password had **expired**.
- **Impact:** loss of **root SSH access** to the VCF Operations appliance. The appliance itself stayed up and serving (uptime 34 days at recovery). **No service/workload impact**.
- **Recurrence exposure:** a VCF Operations alert ("**Management Lifecycle — LCM Root password expiry**," WARNING, active since **2026-07-21**) shows another appliance root is approaching/at expiry, and **vCenter root is known to expire 2026-08-10**.

2. Timeline

Time (approx.)	Event
~appliance deploy	Root password set; a password max-age (expiry) policy was in effect (value not captured — see §5)
2026-07-21	VCF Operations raises " Management Lifecycle — LCM Root password expiry " (WARNING) — early indicator, not acted on
2026-07-27	Automated (<code>plink -batch</code>) SSH to <code>.77</code> root fails as <code>Configured password was not accepted</code>
2026-07-27	Diagnosed as expired (not wrong); changed interactively ; access restored
2026-07-27	Post-fix <code>chage -l root</code> on <code>.77</code> = " Password expires: never "

3. Symptoms

- Non-interactive SSH (`plink -batch` , scripts): **FATAL ERROR: Configured password was not accepted** .

- The appliance is otherwise **up and healthy** (long uptime; services running).
- A standing VCF Operations alert on the **Management Lifecycle** object: **LCM Root password expiry** (impact = HEALTH).

4. Investigation & Evidence (verified)

Check	Method	Result
Appliance alive?	uptime after login	✅ 34 days — the box never went down; only auth was blocked
Wrong vs expired?	interactive login	✅ Expired — an interactive session presented the forced-change flow; entering the same password + a new one restored access
Post-fix expiry state	<code>chage -l root</code> on <code>.77</code>	✅ "Password expires: never" — the fix cleared the expiry
Fleet indicator	VCF Ops <code>GET /suite-api/api/alerts</code>	⚠️ "LCM Root password expiry" active since 2026-07-21 (see §8)
vCenter root	VCF Ops Passwords UI (earlier)	⚠️ Expires 2026-08-10 (change already scheduled)

5. Root Cause

Confirmed root cause: the appliance's **OS root password reached its max-age expiry**. Standard Linux/PAM behavior on an expired password is to **reject non-interactive/password authentication** and to **force a password change on interactive login** — so a scripted `plink -batch` connection reports **"password not accepted"** even though the stored password value is correct. The appliance and its services are unaffected; only credential-based SSH is blocked.

Confidence — honest: - The **expiry as root cause is CONFIRMED** for `.77` — the interactive forced-change flow appeared and resolved it, and the post-fix `chage` shows the expiry cleared. - The **exact original max-age value is NOT stated** — `chage -l root` was **not captured before** the change, so the specific policy (e.g., days-to-expire) is unknown and is deliberately **not guessed here**.

6. Contributing Factors

- **Misleading error string.** `Configured password was not accepted` is indistinguishable at a glance from a genuinely wrong password — it delayed the correct diagnosis.
- **Non-interactive tooling can't self-recover.** `plink -batch`, automation, and monitors cannot complete the interactive forced-change, so an expiry cascades into every scripted path as an apparent auth failure.
- **No proactive expiry management.** The VCF Operations expiry alert existed from 2026-07-21 but was not acted on before the lockout; there was no standing inventory of appliance-root expiry dates.

7. Resolution (verified on .77)

Change the password **interactively** — a real terminal, or the **VM Web Console** (vCenter → the appliance VM → Launch Web Console): 1. `ssh root@<appliance>` → at "You are required to change your password immediately (password expired)", enter the **expired** password, then a **new** one (must meet complexity **and** differ from the old). 2. If SSH rejects the expired password outright, use the **VM Web Console** to change it. 3. **Verify**: log in with the new password; `chage -l root` → "**Password expires: never**" (or a fresh far-future date). Optionally `chage -M -1 root` to disable max-age.

(Runbook: [\[\[RUNBOOK-VCF-ESXi-vCenter-Root-Password-Change-2026-07-27\]\] §3a.](#))

8. Scope / Fleet Exposure

This is a fleet pattern, not a one-off. Appliance roots across the estate share the same aging behavior: - **VCF Operations (.77) root** —  fixed, now **never expires**. - **LCM / Fleet appliance root (.95 / lcm.lab.local)** — the source of the "LCM Root password expiry" alert; **confirmed still valid, expiry ~5 days out (≈2026-08-01)** — **NOT yet expired**. This is the **ideal case**: a **proactive-change window before** any lockout (unlike .7). Change it before it expires — normal interactive `passwd`, then `chage -M -1 root` to disable max-age. (An earlier automated probe hung on an uncached SSH host key, not on expiry — the root is still fully accessible.) - **vCenter root** — **expires 2026-08-10** (verified); change **scheduled 2026-08-08**. - **SDDC Manager root** — expiry **not verified**; include in the audit below.

9. Preventive Actions / Recommendations

1. **Build a one-time expiry inventory** — `chage -l root` on **every** management appliance (VCF Ops, LCM/Fleet, vCenter, SDDC Manager, NSX). Record each expiry date.
2. **Remediate the currently-flagged root(s)** — the LCM/Fleet appliance (§8), interactively.
3. **Prevent recurrence** — either **rotate proactively before expiry** on a schedule, or **disable max-age** (`chage -M -1 root`) where the security policy permits (as was effectively done on .77).
4. **Watch the signal** — treat the VCF Operations "LCM Root password expiry" alert as an actionable early-warning, not background noise.
5. **Document each change** — new values to the vault; note that appliance roots are **separate** from each other and from any lab-wide credential.

10. How to Diagnose This Quickly (for next time)

- **Symptom pattern**: SSH says "password not accepted" **but** the appliance is clearly up and the password is believed correct → **suspect expiry first**, not a wrong password.
- **Confirm**: log in **interactively** (a forced-change prompt = expired), or once in run `chage -l root`, or check the VCF Operations expiry alert.
- **Do NOT** assume the password was changed/lost and start credential-reset detours — check expiry first.

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Revision History

Version	Date	Notes
1.0	2026-07-27	Initial RCA. VCF Operations (.77) root SSH lockout root-caused to an EXPIRED root password (presented as "password not accepted"); confirmed via the interactive forced-change fix + post-fix chage. Fleet exposure documented; original max-age not captured, so not stated.
1.1	2026-07-27	Corrected §8: the Fleet/LCM (.95) root is NOT expired — it expires in ~5 days (≈2026-08-01), a proactive-change window (verified by operator). The earlier automated probe hung on an uncached SSH host key, not on expiry. Removed the incorrect "expired-blocks-SSH" inference for .95.

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